Capstone Exploratory Data Analysis

Dain Lee, Rameena Quddus, and Sloka Jagannathan

Arizona State University, Tempe, AZ 85281, USA

October 29, 2025

1 Data Sources, Preparation, and Cleaning

This study draws on two primary datasets: `cleaned_makeup_reviews.csv` and `cleaned_makeup_products.csv` [[2], [1]]. Together, these provide a detailed view of consumer feedback and product attributes in the beauty industry. The reviews dataset contains 314,029 entries and 27 features, each representing an individual customer review. It includes unique identifiers, review content and ratings, engagement data such as helpful votes, reviewer nicknames, and timestamps for when each review was created and updated. The products dataset complements this by providing product-level information, including brand, category, price, and related metadata.

Before conducting any analysis, both datasets were reviewed for consistency, accuracy, and completeness. Data types were standardized to ensure compatibility across columns and between datasets. Duplicate records were identified and removed based on unique identifiers to prevent repeated entries from skewing descriptive statistics. Records containing invalid or missing values were handled appropriately to preserve data quality. For numerical columns, missing values were replaced with zeros to maintain analytical continuity, while categorical fields were carefully inspected to confirm valid classifications. All date fields were reformatted into a consistent structure, enabling accurate time-series comparisons and temporal trend analyses.

````
```{r}
# Changing all null values from Makeup Reviews data set into 0 (numerical) or Unknown (categorical)

makeup_reviews <- lapply(makeup_reviews, function(x) {
  if (is.character(x) || is.factor(x)) {
    x[is.na(x)] <- "Unknown"
  } else if (is.numeric(x)) {
    x[is.na(x)] <- 0
  } else if (is.logical(x)) {
    x[is.na(x)] <- FALSE
  }
  return(x)
}) |> as.data.frame()

```
````

Figure 1: Modifying null values in Makeup Reviews dataset.

```{r}
# Changing all null values from Makeup Products data set into 0 (numerical) or Unknown (categorical)

makeup_products <- lapply(makeup_products, function(x) {
  if (is.character(x) || is.factor(x)) {
    x[is.na(x)] <- "Unknown"
  } else if (is.numeric(x)) {
    x[is.na(x)] <- 0
  } else if (is.logical(x)) {
    x[is.na(x)] <- FALSE
  }
  return(x)
}) |> as.data.frame()
```

Figure 2: Modifying null values in Makeup Products dataset.

```{r}
# Checking null values in Makeup Review data set afterwards
sum(is.na(makeup_reviews))
```

```
[1] 0
```

```{r}
#Checking new number of null values in Makeup Products data set (should be 0)
sum(is.na(makeup_products))

```

```
[1] 0
```

Figure 3: Checking the updated number of null values in both data sets.

Text preprocessing involved normalizing capitalization, removing special characters, HTML tags, and other unwanted formatting artifacts. This step ensured that product descriptions, review text, and brand names were clean and uniform for subsequent natural language processing tasks. Rating values were verified to fall within the valid 1–5 range, and an additional derived variable, the helpfulness ratio, was computed to quantify user engagement by measuring the proportion of helpful votes relative to total votes.

During data cleaning, several inconsistencies were discovered in the `cleaned_makeup_products.csv` [1] file, particularly in the `brand` field. Certain brand names were misspelled, truncated, or entirely missing due to formatting errors during export from Excel. For example, some entries such as "Lancome" appeared as "Lancom" or were omitted altogether. To address this, a hybrid correction approach was used: minor typographical inconsistencies were fixed using automated find-and-replace operations, while missing or ambiguous brands were manually cross-checked against product names and reentered. Following these corrections, frequency counts were used to verify that brand names were consistent and aligned across all records. Columns that would not be used for further analysis in the project were dropped.

```
```{r}
#Dropping unnecessary columns in both data sets

# Drop product_link and product_link_id columns in makeup_products
makeup_products_clean <- makeup_products %>%
  select(-c(product_link, product_link_id))

# Drop legacy_id, ugc_id, and uri columns in makeup_reviews
makeup_reviews_clean <- makeup_reviews %>%
  select(-c(legacy_id, ugc_id, uri))

```
```

Figure 4: Dropping unnecessary columns in both datasets.

Finally, exploratory checks such as summary statistics and outlier detection were conducted to confirm the integrity of the cleaned data. These preprocessing steps were essential for ensuring that subsequent analyses on customer sentiment, product performance, and engagement patterns were based on accurate and reliable data. The cleaned datasets contained 1373 unique rows for makeup products and 314,029 rows for makeup reviews.

```
Rows: 1,373
Columns: 33
$ category
$ item_id
$ product_name
$ brand
$ price
$ num_shades
$ rating
$ num_reviews
$ description
$ pros
$ cons
$ best_uses
$ describe_yourself
$ review_star_1
$ review_star_2
$ review_star_3
$ review_star_4
$ review_star_5
$ rating_star_1
$ rating_star_2
$ rating_star_3
$ rating_star_4
$ rating_star_5
$ rating_count
$ review_count
$ average_rating
$ recommended_ratio
$ native_review_count
$ native_sampling_review_count
$ native_community_content_review_count
$ syndicated_review_count
$ faceoff_negative
$ faceoff_positive
```

```
Rows: 314,029
Columns: 24
$ unique_review_id
$ product_link_id
$ review_id
$ type
$ id
$ internal_review_id
$ headline
$ nickname
$ created_date
$ updated_date
$ rating
$ helpful_votes
$ not_helpful_votes
$ comments
$ locale
$ location
$ bottom_line
$ product_page_id
$ upc
$ gtin
$ is_staff_reviewer
$ is_verified_buyer
$ is_verified_reviewer
$ helpful_score
```

Figure 5: Makeup reviews and products dataset variables.

2 Exploratory Data Analysis

2.1 Distribution of Individual Review Ratings

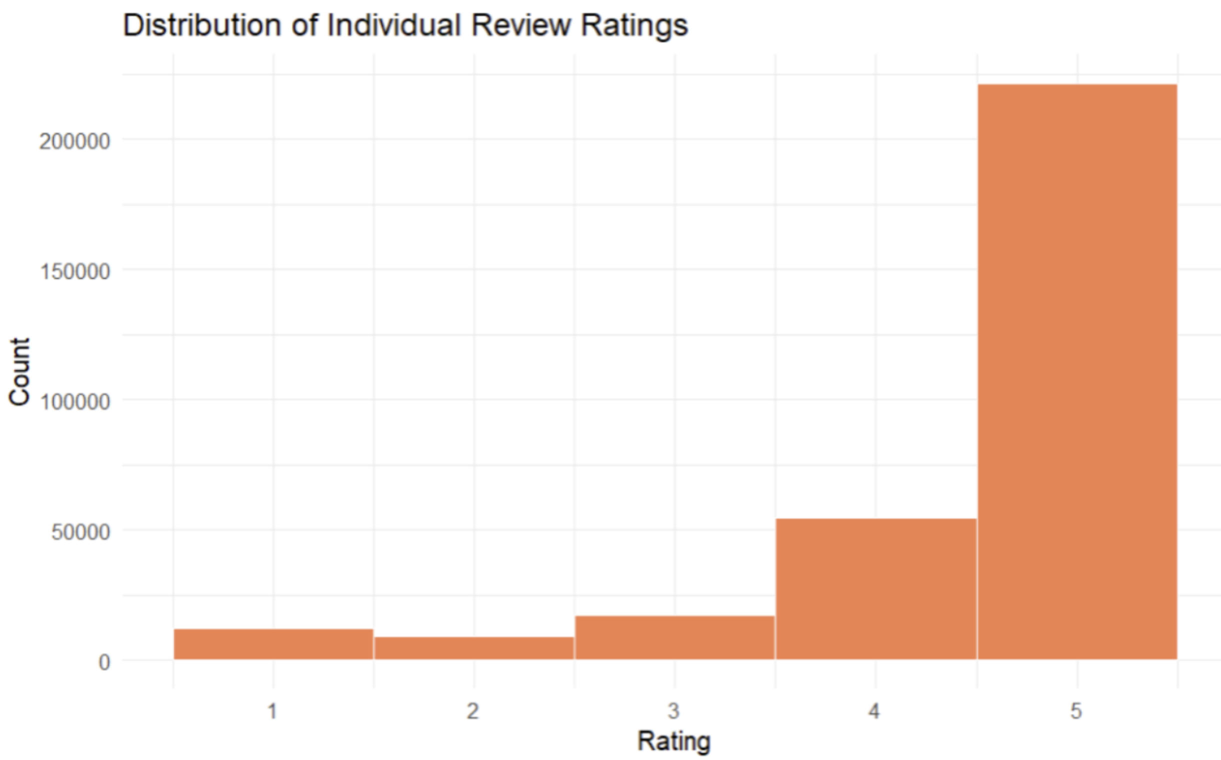

Figure 6: Distribution of customer ratings for makeup products.

This bar chart displays the distribution of individual review ratings across all makeup products. The results reveal a clear skew toward higher ratings, with 5-star reviews making up the overwhelming majority. A smaller but noticeable portion of reviews are rated 4 stars, while 1-, 2-, and 3-star ratings occur far less frequently. This distribution suggests that customers generally have very positive experiences with the products they purchase. However, it also indicates a potential positive bias, where users who are satisfied are more likely to leave feedback than those who are neutral or dissatisfied. Overall, this visualization emphasizes the strong customer approval of the products, though future analyses should consider this imbalance when interpreting sentiment or rating-based insights.

2.2 Price Distribution by Top 10 Brands

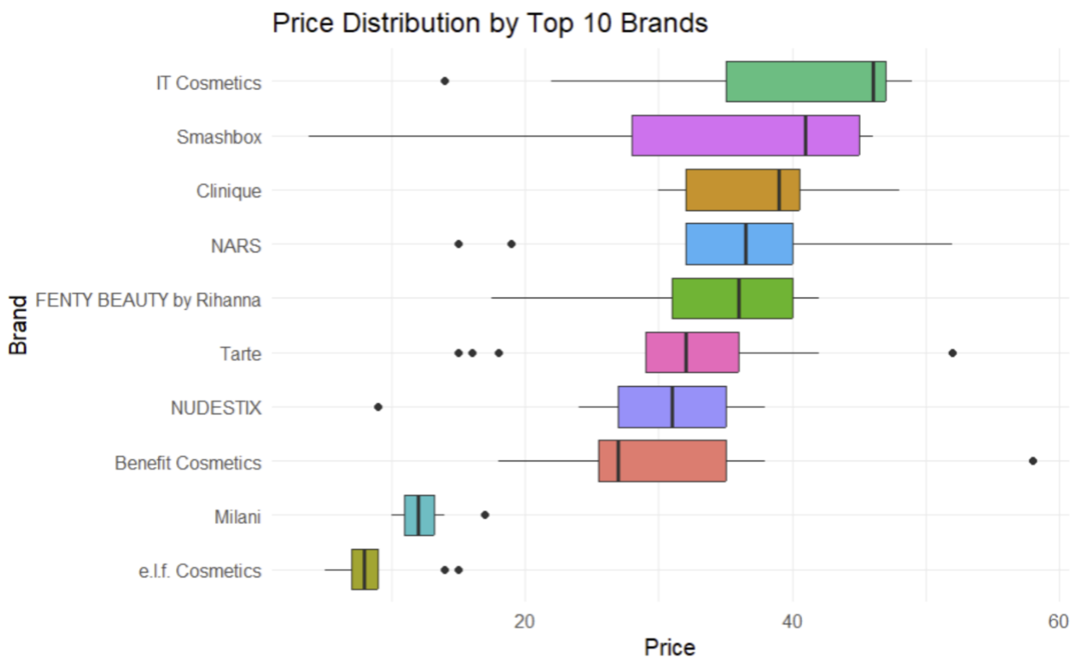

Figure 7: Price distribution across Top 10 makeup brands.

This boxplot visualizes the price distribution across the top 10 makeup brands, offering a clear comparison of pricing ranges and variability between brands. The chart shows that IT Cosmetics, Smashbox, and NARS tend to have higher median prices, suggesting they target more premium market segments. Clinique, Fenty Beauty by Rihanna, and Tarte fall into the mid-range category with moderately priced products and relatively consistent pricing. On the other hand, Milani and e.l.f. Cosmetics have noticeably lower price ranges, indicating that these brands cater to more budget-conscious consumers. The varying box widths and presence of outliers indicate differences in product variety and pricing strategies, brands like Smashbox and Benefit Cosmetics show wider spreads, implying diverse product offerings across price points. Overall, this visualization highlights clear segmentation in the beauty market, with distinct tiers of luxury, mid-range, and affordable brands. This provides valuable context for understanding how pricing aligns with brand positioning and consumer perception.

2.3 Review Volume Over Time

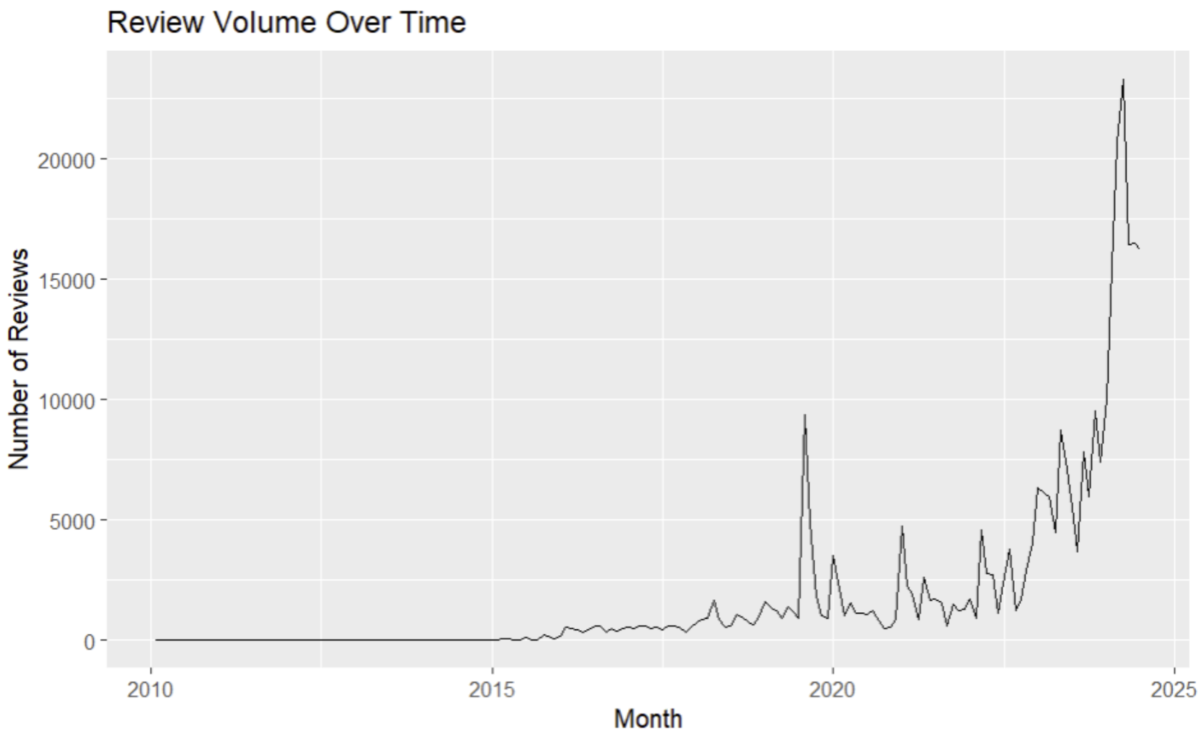

Figure 8: Trend of makeup product review volume from 2010 to 2025.

This line graph illustrates the trend of review volume over time for makeup products from 2010 to 2025. The data reveal a relatively low and stable number of reviews in the early years, followed by a noticeable rise starting around 2018. A sharp spike appears around 2020, which may correspond to increased online shopping activity during the pandemic period, and review activity continues to climb dramatically afterward. The number of reviews peaks sharply near 2024, reaching over 20,000 per month, before showing a slight decline. Overall, the trend demonstrates a significant growth in consumer engagement over time, reflecting both the expansion of the beauty market and the growing reliance on online platforms for product feedback and purchasing decisions.

2.4 Normal Distribution Plots for Makeup Product Numerical Variables

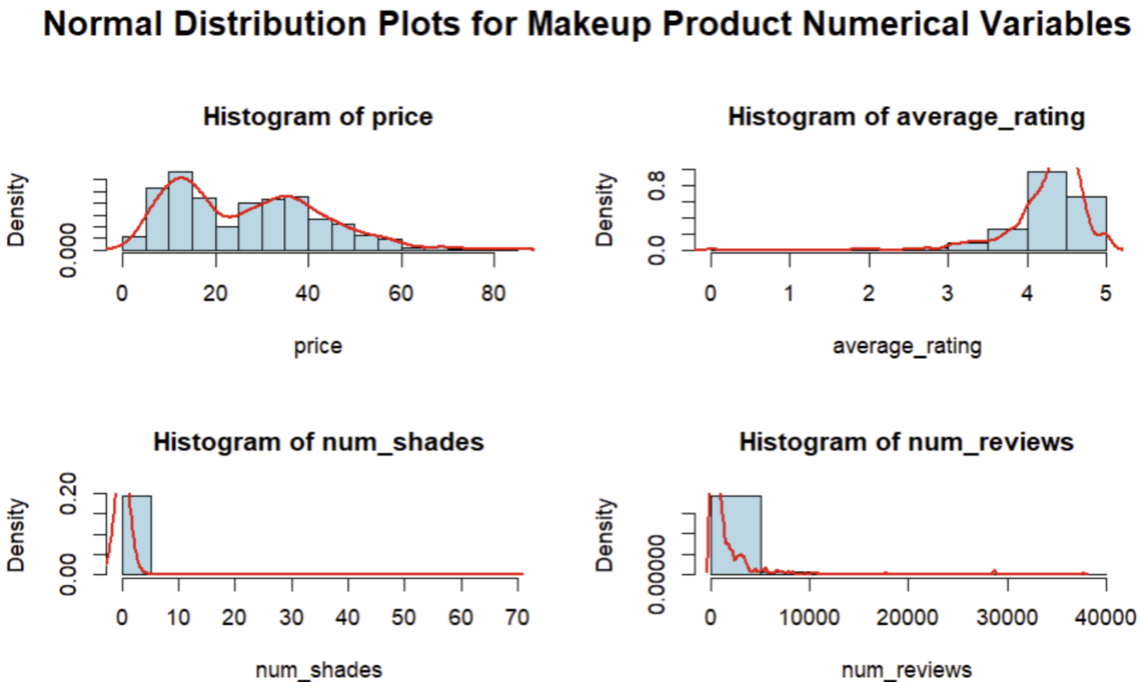

Figure 9: Distribution of key numerical variables for makeup products.

This figure presents histograms with density curves for four numerical variables in the makeup product dataset: price, average rating, number of shades, and number of reviews. The price distribution is slightly right skewed, showing that most products are moderately priced while a smaller number are much more expensive. The average rating distribution is left skewed, with most products receiving high ratings between 4 and 5, which aligns with the earlier finding of generally positive customer feedback. Both the number of shades and the number of reviews show strong right skewness, meaning that only a few products have a very large variety of shades or a very high number of reviews, while most have relatively few. Overall, these distributions indicate that the data is not normally distributed, especially for count based variables, and that there is noticeable variation in product popularity and diversity within the makeup market.

2.5 Distribution Shape Analysis of Key Variables

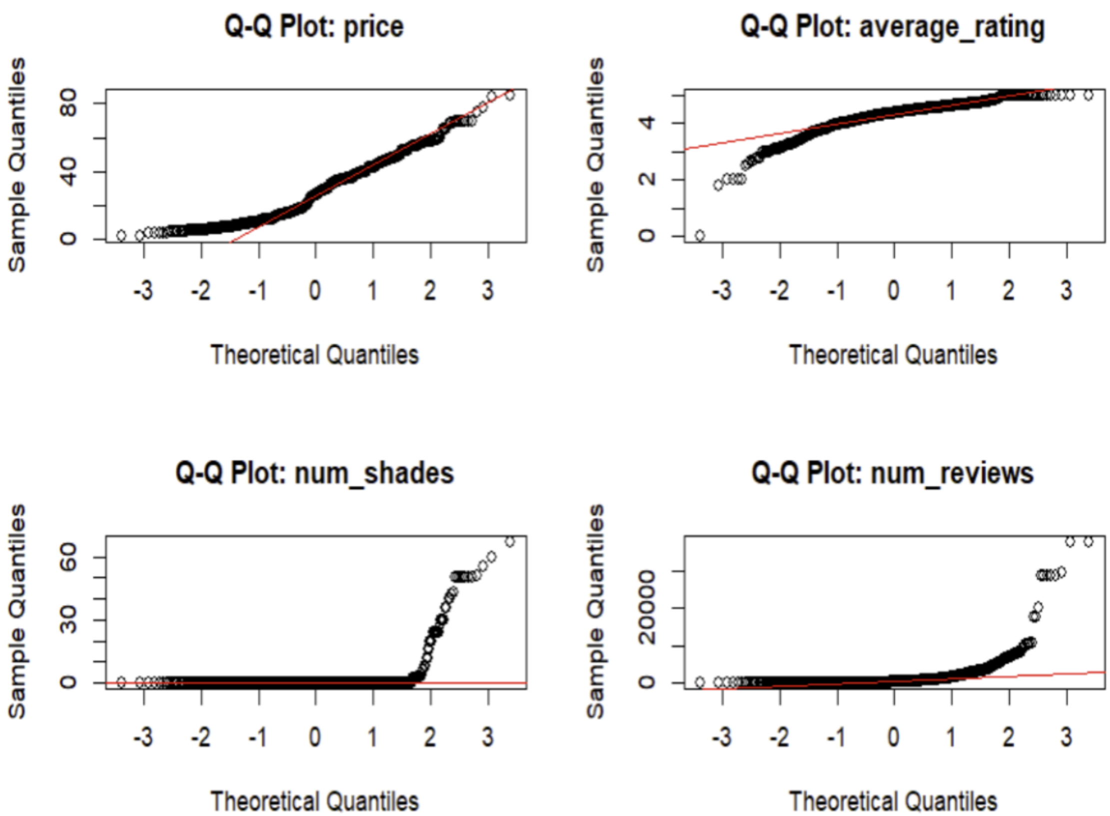

Figure 10: Q-Q Plots assessing normality of makeup product variables.

These Q-Q plots assess whether the numerical variables in the makeup product dataset follow a normal distribution. In all four plots, the data points deviate notably from the diagonal red line, indicating that none of the variables are normally distributed. The plot for price shows moderate deviation at both tails, suggesting slight skewness with some high-priced outliers. The average_rating plot is left skewed, with most values concentrated near the upper limit, reflecting generally high ratings. Both num_shades and num_reviews display strong upward curvature, revealing extreme right skewness caused by a few products having an exceptionally large number of shades or reviews compared to most others. Overall, these Q-Q plots confirm that the dataset contains several non-normal variables with skewed distributions, meaning transformations or nonparametric methods may be more appropriate for further statistical analysis.

2.6 Normal Distribution Plots for Makeup Review Numerical Variables

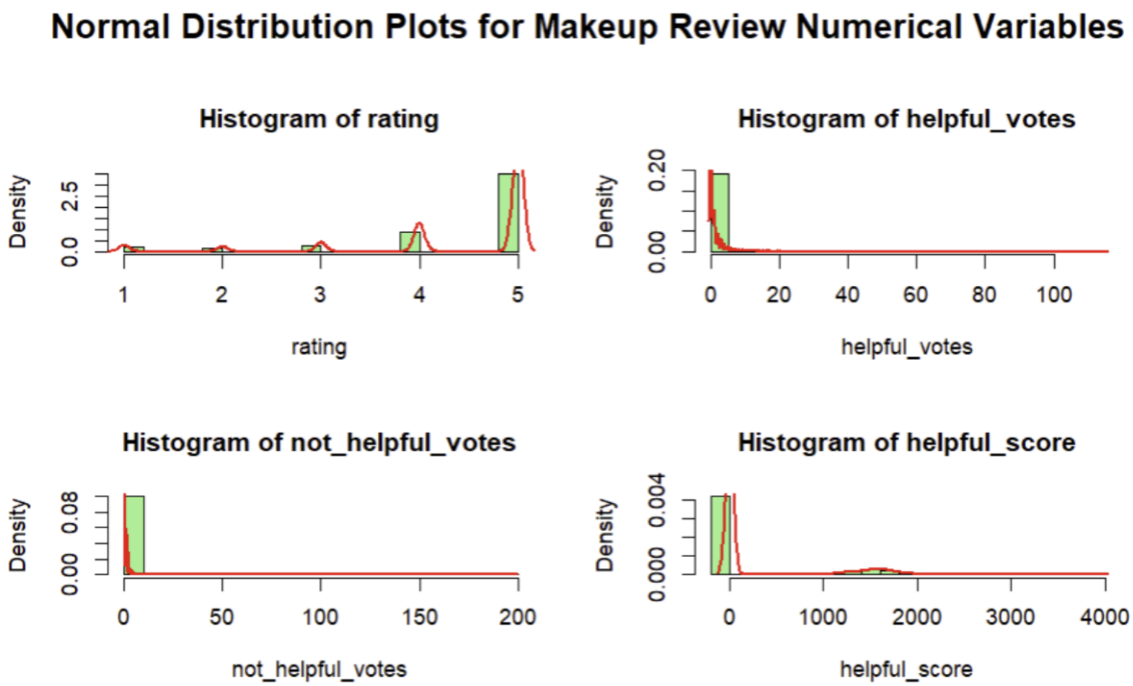

Figure 11: Distribution of makeup review metrics with normal density curves.

The histograms above display the distribution of four key numerical variables from the makeup reviews dataset: rating, helpful_votes, not_helpful_votes, and helpful_score, with overlaid normal density curves in red. The rating variable shows a strong right skew, with most reviews rated between 4 and 5, indicating predominantly positive feedback. Both helpful_votes and not_helpful_votes are heavily right skewed, suggesting that the majority of reviews receive few to no votes, while a small number of reviews attract many. Similarly, helpful_score displays a sharp peak near zero with a long right tail, indicating that only a limited number of reviews receive high helpfulness scores. Overall, none of the variables follow a normal distribution, and all show significant skewness and kurtosis which is typical of user generated review data.

2.7 Normality Check of Review Metrics

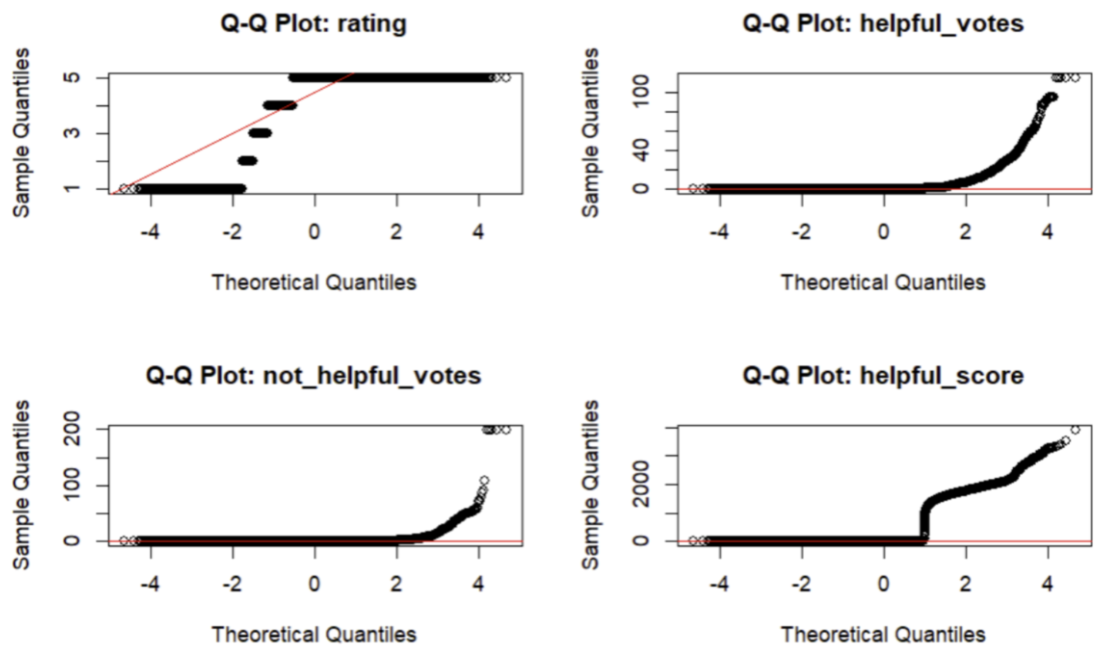

Figure 12: Q-Q Plots assessing normality of makeup review variables.

The Q-Q plots above display the distribution of the numerical variables rating, helpful_votes, not_helpful_votes, and helpful_score compared to a theoretical normal distribution. In all four plots, the data points deviate substantially from the red reference line, confirming that none of these variables follow a normal distribution. The rating variable shows strong clustering near the upper end, with most values concentrated around 4 and 5, indicating skewness toward higher ratings. The helpful_votes, not_helpful_votes, and helpful_score variables show heavy right skewness, where most observations are near zero and a few extreme values lie far above the expected line. These patterns suggest the presence of outliers and long tails in the data, which are common in user-generated review datasets.

2.8 Correlation Matrix of Makeup Reviews

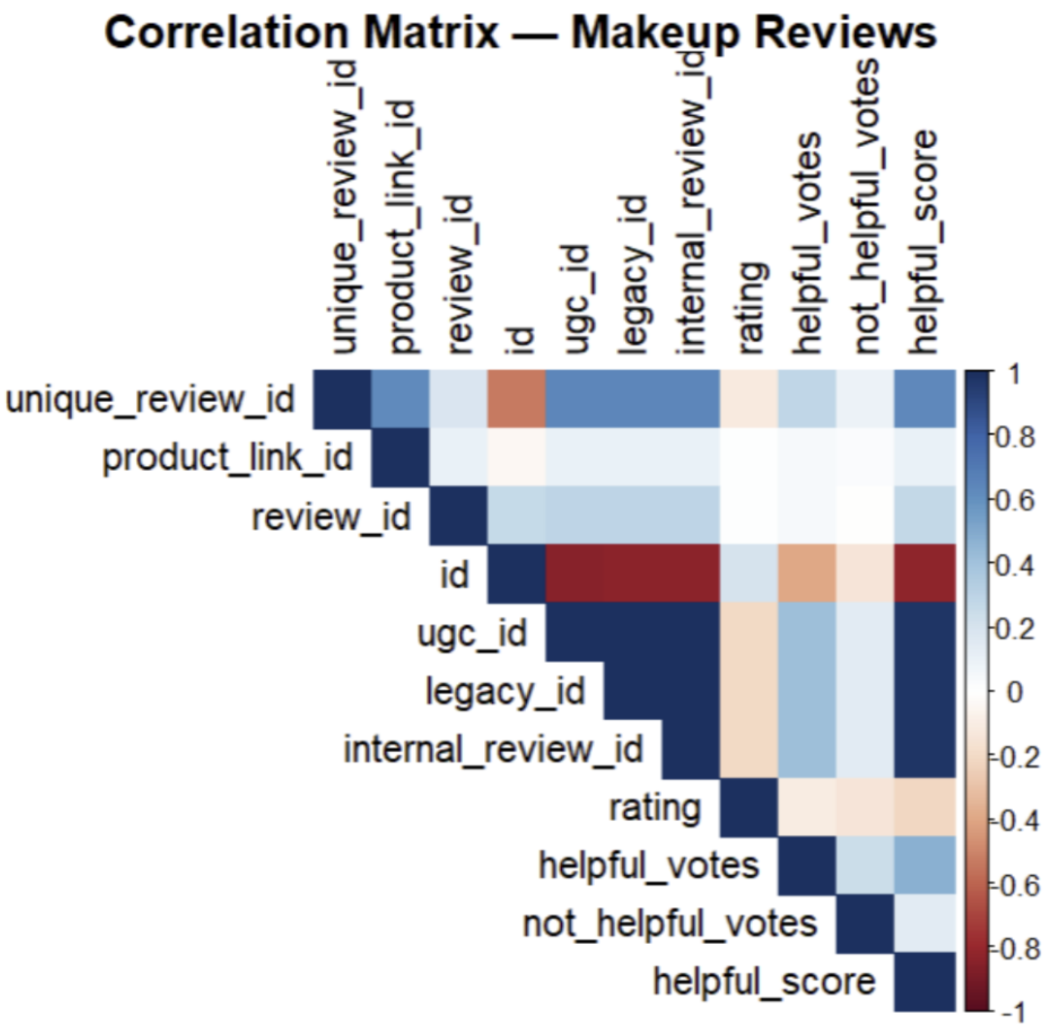

Figure 13: Correlation matrix of key variables in makeup review dataset.

This correlation matrix visualizes the relationships among key numerical variables in the makeup reviews dataset. Most ID-based variables, such as unique_review_id, product_link_id, and review_id, show weak or no meaningful correlation with other variables since they serve as identifiers rather than data features. However, there are stronger positive correlations observed among engagement-related variables like helpful_votes, not_helpful_votes, and helpful_score, suggesting that reviews receiving more interactions tend to have higher helpfulness scores. The variable rating shows only a weak correlation with helpfulness measures, indicating that high ratings do not necessarily correspond to higher helpful votes. Overall, this matrix highlights that while identification fields are largely independent, user interaction metrics share moderate positive relationships that may be useful for understanding review credibility or engagement patterns.

2.9 Correlation Matrix of Makeup Products

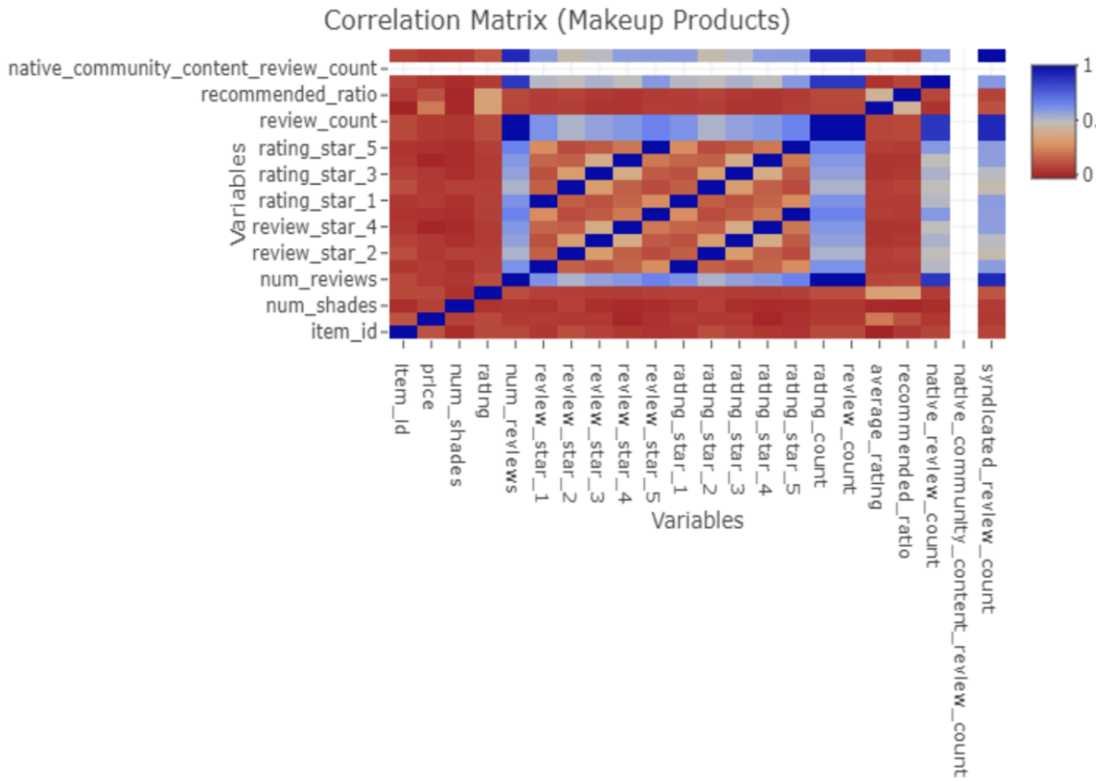

Figure 14: Correlation matrix of makeup product features and engagement metrics.

This correlation matrix for makeup product variables shows the relationships between various numerical and categorical features in the dataset. The darker blue areas indicate stronger positive correlations, while lighter shades and reddish tones represent weaker or negative relationships. Several groups of variables, particularly those related to review counts (such as rvw__1 through rvw__5) and return or rating components (rtn__1, rtn__2, etc.), exhibit strong internal correlations, suggesting that they measure similar aspects of product performance or user engagement. Meanwhile, variables like price, num_shades, and rating show weaker correlations with other attributes, implying that pricing, product variety, and user ratings operate somewhat independently. Overall, this visualization highlights that the dataset contains clusters of interrelated metrics, especially within review and rating features, while core product characteristics such as price and shade diversity remain largely distinct from engagement-based measures.

2.10 Makeup Products Distributions

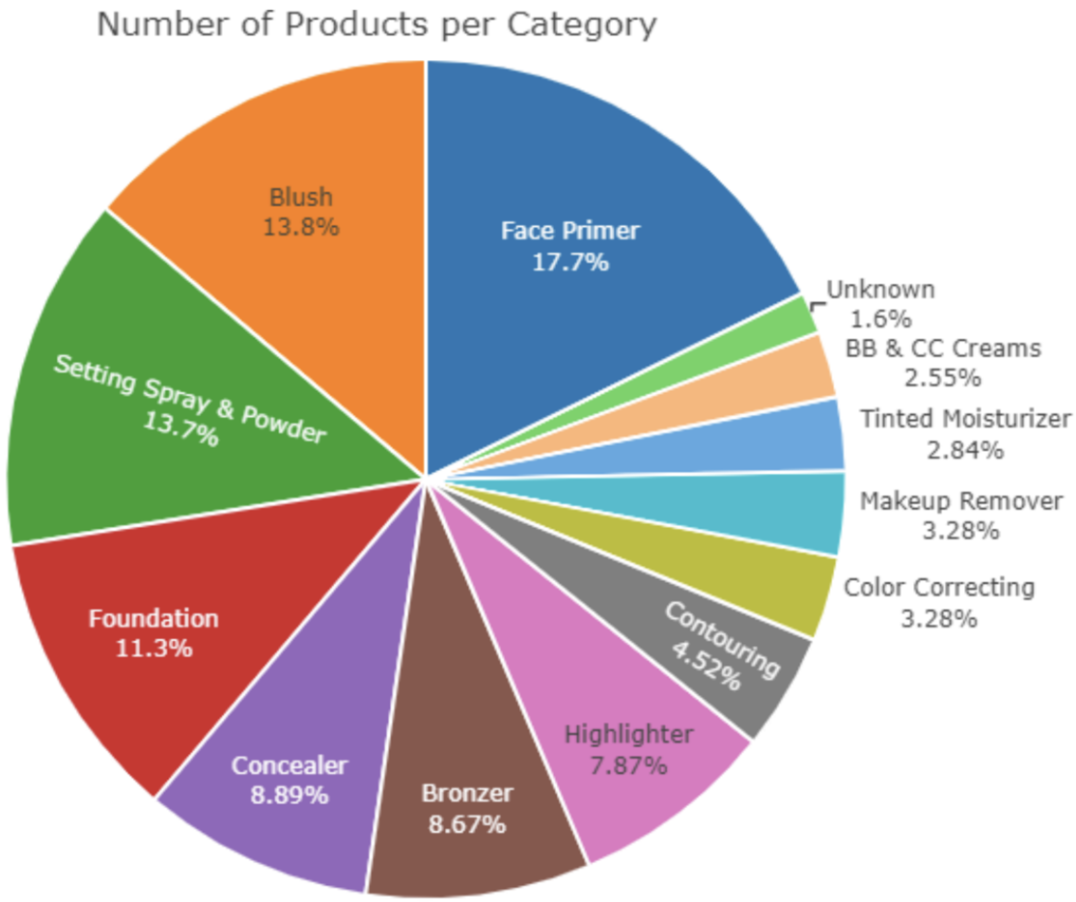

Figure 15: Distribution of makeup products by category.

The pie chart illustrates the distribution of makeup products across various categories. Face Primer represents the largest proportion, accounting for 17.7 percent of all products, followed by Blush at 13.8 percent and Setting Spray and Powder at 13.7 percent. Foundation makes up 11.3 percent, while Concealer and Bronzer contribute 8.9 percent and 8.7 percent respectively. Other notable categories include Highlighter at 7.9 percent and Contouring at 4.5 percent. Smaller portions are observed for Color Correcting, Makeup Remover, Tinted Moisturizer, and BB and CC Creams, each representing less than 4 percent of the total. A very small portion, 1.6 percent, falls under the Unknown category. Overall, the chart indicates that the dataset includes a broad range of product types, with a concentration in essential base products such as primers, blushes, and setting powders.

2.11 Distribution of Product Prices

Figure 16: Distribution of makeup product prices across price ranges.

The bar chart shows the distribution of makeup product prices across different price ranges. Most products are priced between 10 and 20 dollars, making this the most common price range. A notable number of products also fall within the 30 to 40 dollar range, followed by smaller but still considerable groups in the 0 to 10 and 20 to 30 dollar ranges. The number of products decreases steadily as prices increase beyond 40 dollars, with very few items priced above 70 dollars. Overall, the chart indicates that the majority of products are affordably priced, suggesting that most items in the dataset belong to mid-range or budget-friendly categories rather than luxury-priced products. There also appears to be a greater concentration of lower average ratings for products in the lower price ranges, particularly between 0 and 20 dollars, compared to those priced between 30 and 60 dollars.

2.12 Relationship Between Price and Rating

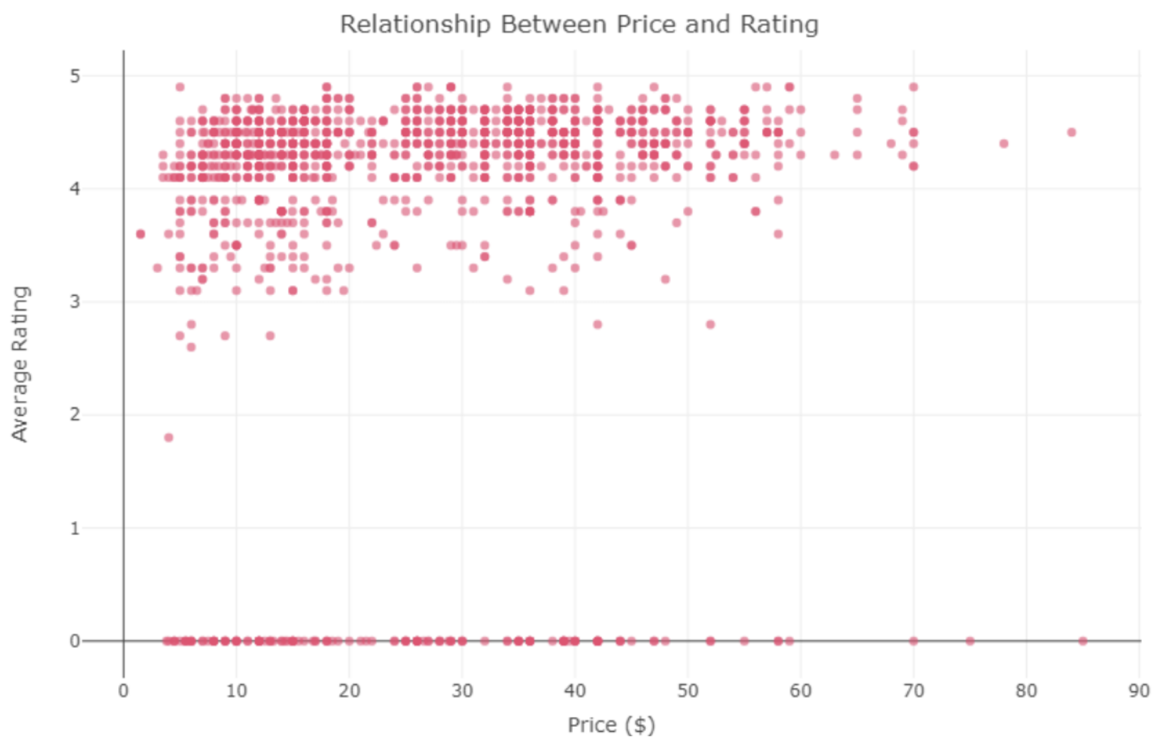

Figure 17: Relationship between makeup product price and customer rating.

The scatterplot displays the relationship between the price of the product and the average rating of of makeup products. Overall, there is no strong linear relationship between the two variables, as high ratings are observed across nearly all price ranges. Most products, regardless of price, cluster between ratings of 4 and 5, indicating generally positive customer feedback. However, lower-priced products, particularly those under 20 dollars, show a wider spread of ratings, including several lower ratings, suggesting greater variability in quality perception. In contrast, mid-range and higher-priced products tend to maintain consistently high ratings with fewer low outliers. This pattern suggests that while price alone does not determine customer satisfaction, higher-priced products may be perceived as more reliable or of better quality.

2.13 Relationship Between Number of Shades and Product Ratings

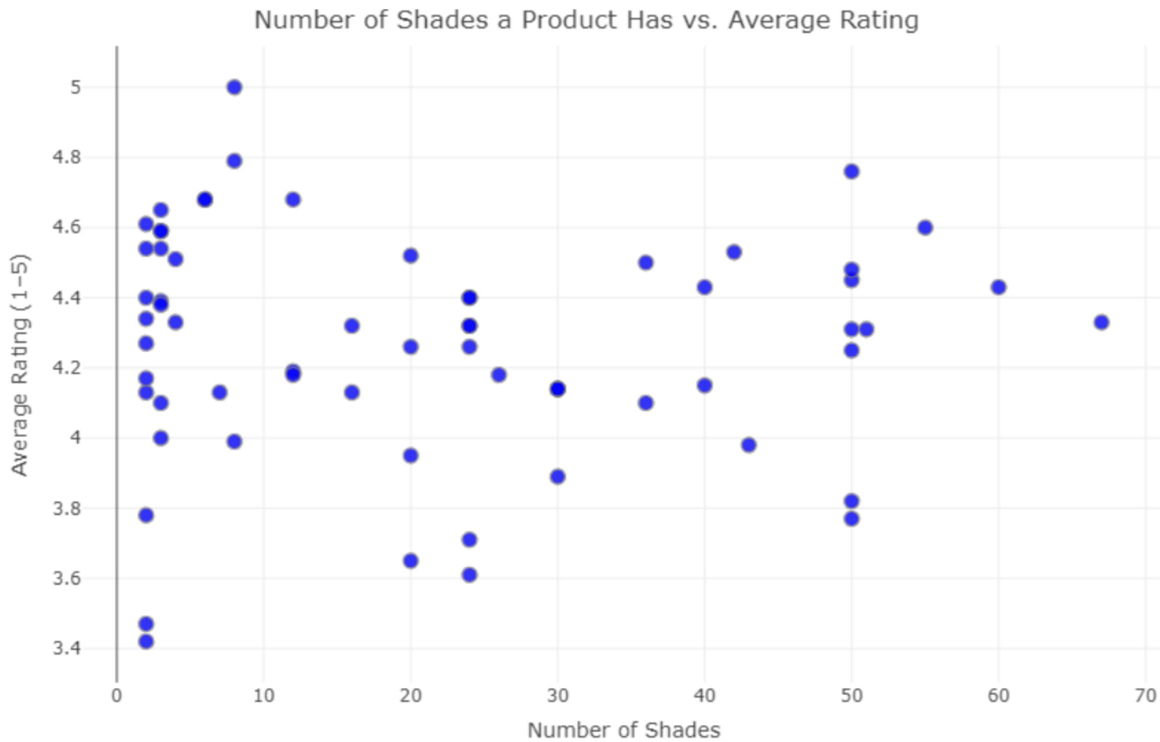

Figure 18: Relationship between number of shades and average product rating among makeup products.

The scatterplot illustrates the relationship between the number of shades a product offers and its average rating. Tarte Amazonian Clay 16-hour Full Coverage Foundation and KVD Beauty Good Apple Lightweight Full-Coverage Concealer have 2 shades, and are the 2 lowest-ranked products in this graph. There is a slight trend of products with a larger number of shades being higher-ranked. However, for the most part, products with both few and many shades tend to receive similar ratings. Most products have average ratings between 4.0 and 4.5, regardless of how many shades they include. A slight clustering can be observed among products with fewer than 10 shades, suggesting that a majority of items in the dataset have limited shade options. Although a few products with a higher number of shades also receive high ratings, this trend is not consistent enough to suggest that offering more shades directly impacts customer satisfaction. This indicates that while shade variety may contribute to inclusivity and choice, it does not necessarily influence overall product ratings.

2.14 Number of Reviews per Rating Count

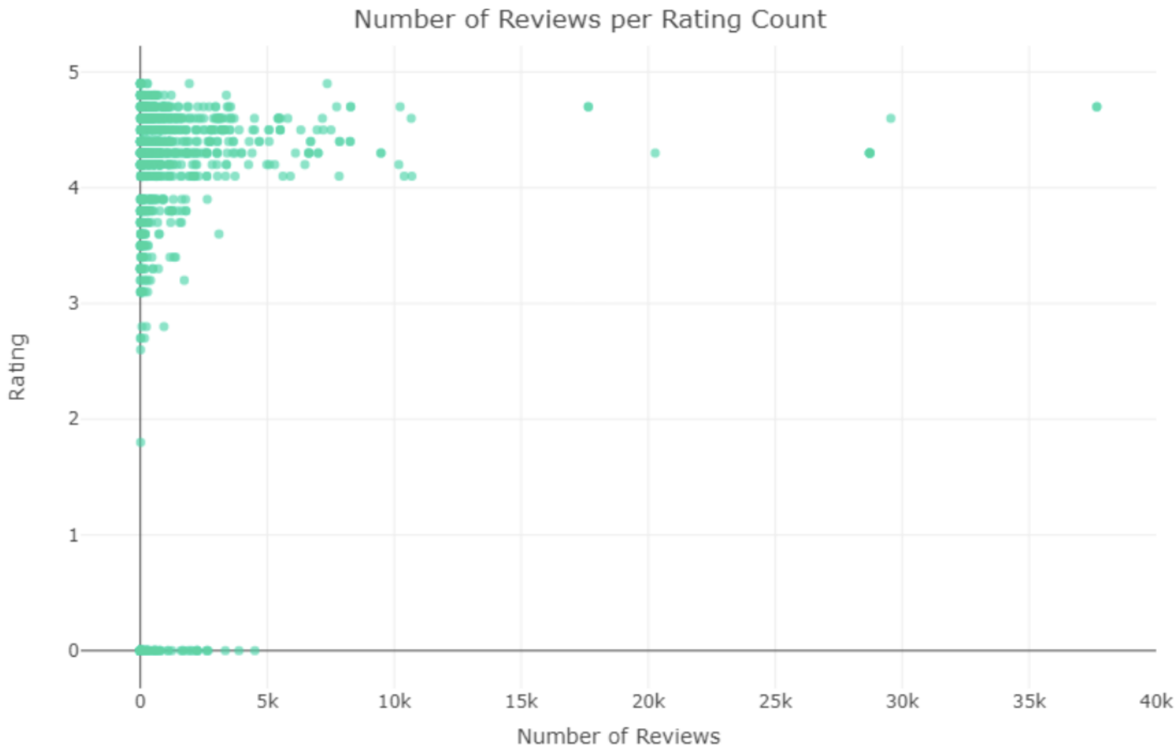

Figure 19: Relationship between number of reviews and product rating.

The scatterplot shows the relationship between the number of reviews a product has received and its average rating. Most products cluster between ratings of 4 and 5, indicating that the majority of reviewed items are well-rated. Products with fewer than 5,000 reviews dominate the dataset, and very few have extremely high review counts. It is uncommon for products to have ratings below 3, suggesting that low-rated items are rare in this dataset. Overall, the figure highlights that while the number of reviews varies widely, most products tend to receive positive feedback regardless of how many reviews they have.

2.15 Correlation of Numerical Variables with Average Product Rating

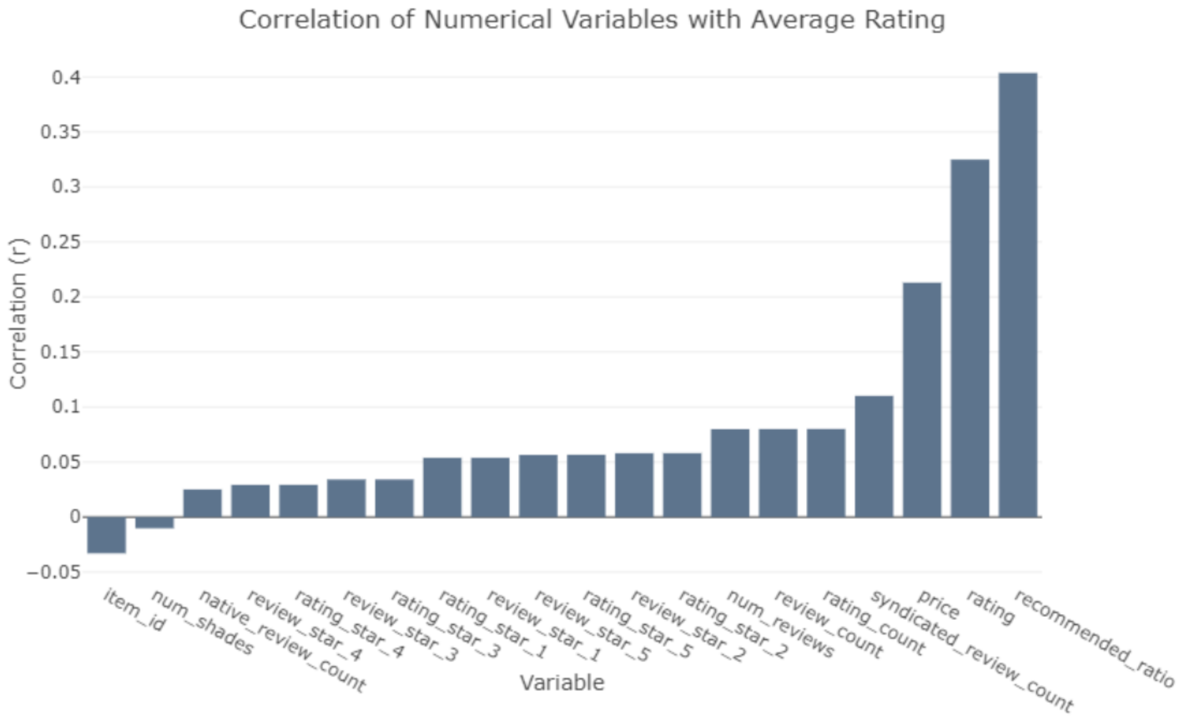

Figure 20: Correlation coefficients between numerical variables and the average product rating.

The bar chart illustrates the correlation coefficients between numerical variables in the makeup product dataset and the target variable, average rating. Variables on the right side of the chart show stronger positive correlations, while those on the left have weaker or slightly negative relationships. The recommended_ratio and syndicated_review_count variables have the highest positive correlations with average rating, suggesting that products more frequently recommended by users or widely reviewed tend to receive higher ratings. Price and rating_count also show moderate positive correlations, indicating that higher-priced products and those with more reviews are slightly more likely to achieve higher average ratings. In contrast, variables such as item_id and num_shades show minimal or no correlation with average rating. Overall, the chart helps identify which features are most relevant for predicting product ratings and can be considered strong candidates for feature importance testing in future machine learning models.

2.16 Brands with Highest Number of Reviews

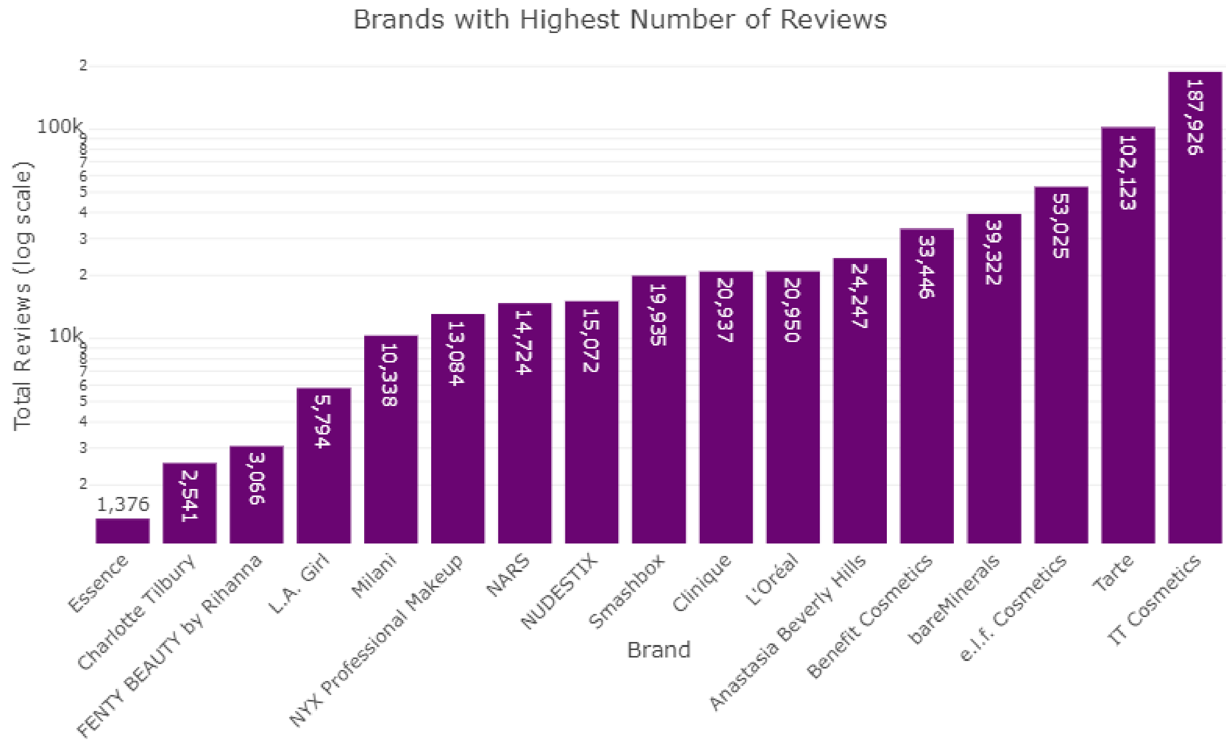

Figure 21: Brands with the highest total number of makeup reviews.

The bar chart presents the brands with the highest total number of reviews in the makeup dataset, displayed on a logarithmic scale. IT Cosmetics has the greatest number of reviews with 187,926 reviews and Tarte with 102,123 reviews. Other highly reviewed brands include e.l.f. Cosmetics, bareMinerals, and Benefit Cosmetics, each exceeding 30,000 reviews. Brands such as Anastasia Beverly Hills, L'Oréal, and Clinique have between 20,000 and 25,000 reviews, while brands such as Charlotte Tilbury and FENTY BEAUTY by Rihanna have under 5,000. Overall, the distribution shows that a few top brands dominate the review counts, suggesting stronger customer engagement or broader market reach in comparison to smaller or niche brands. It also suggests that the distribution of brands represented in the dataset is uneven so results interpreted should take that into consideration.

2.17 Brands with Highest Average Ratings

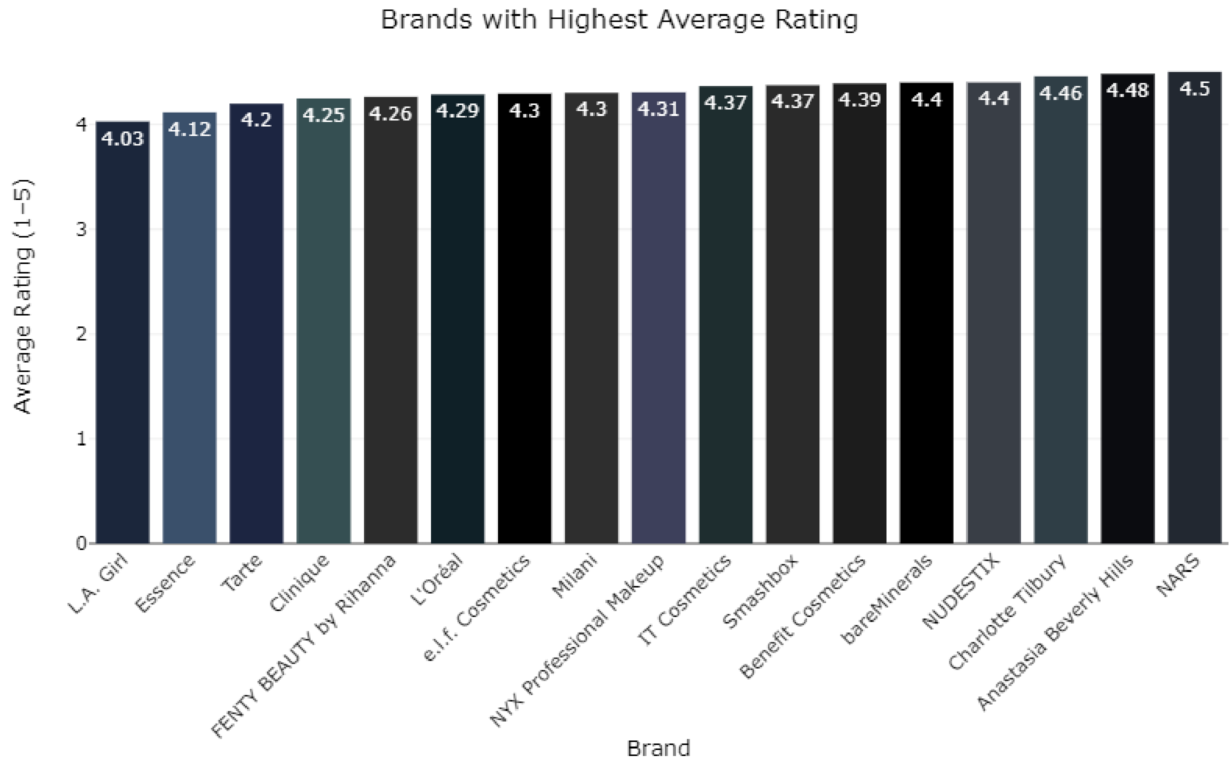

Figure 22: Brands with the highest average makeup product ratings.

The bar chart displays the brands with the highest average product ratings in the makeup dataset. All listed brands maintain high ratings, ranging from approximately 4.18 to 4.5 on a 5-point scale, indicating overall strong customer satisfaction. NARS holds the highest average rating at 4.5, closely followed by Anastasia Beverly Hills and Charlotte Tilbury, both above 4.45. Other well-rated brands include NUDESTIX, Benefit Cosmetics, and Smashbox, each with average ratings above 4.3. All of these brands maintain an average rating above 4.0 across their various products. Overall, the chart shows that the top makeup brands in this dataset consistently receive positive feedback, suggesting strong product quality and customer approval across the board.

2.18 Products with Highest Average Ratings

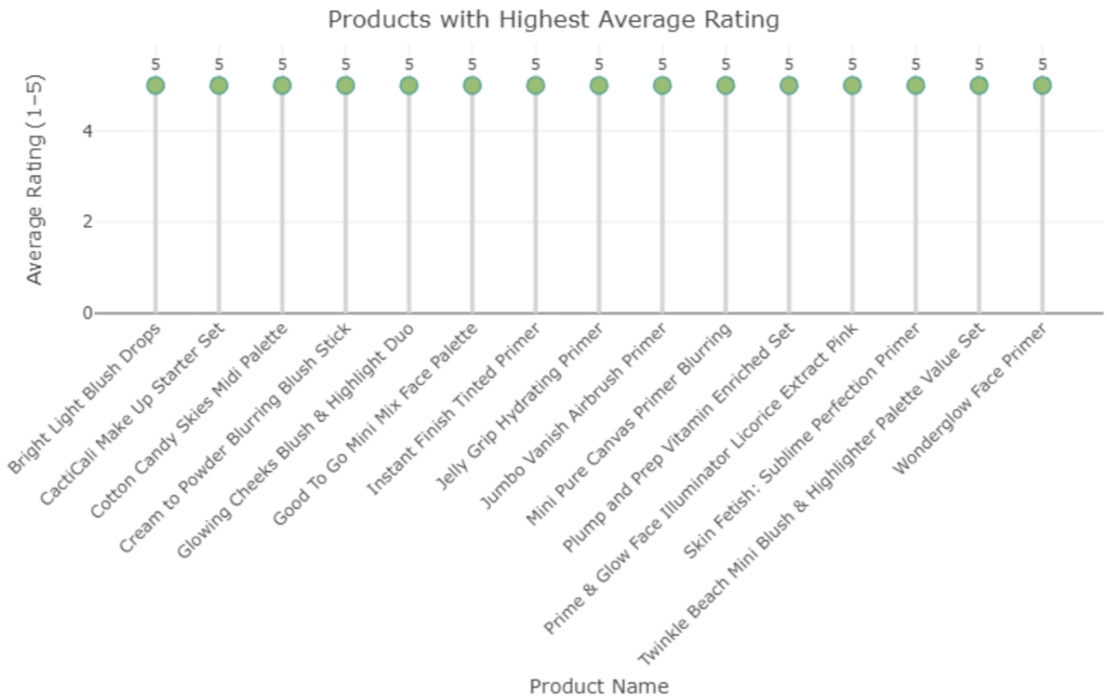

Figure 23: Top makeup products with perfect customer ratings.

The bar chart highlights the makeup products with the highest average ratings in the dataset. Each of the listed products has a perfect average rating of 5, reflecting extremely positive customer feedback. These top-rated items include a mix of primers, blushes, palettes, and highlighting products such as the Bright Light Blush Drops, Jelly Grip Hydrating Primer, Wonderglow Face Primer, and Twinkle Beach Mini Blush and Highlighter Palette Value Set. The consistency of perfect ratings across various brands and product types suggests strong consumer satisfaction and high product performance. However, it may also indicate a smaller sample size for some products, as perfect scores can occur when there are few total reviews.

2.19 Products with Lowest Average Ratings

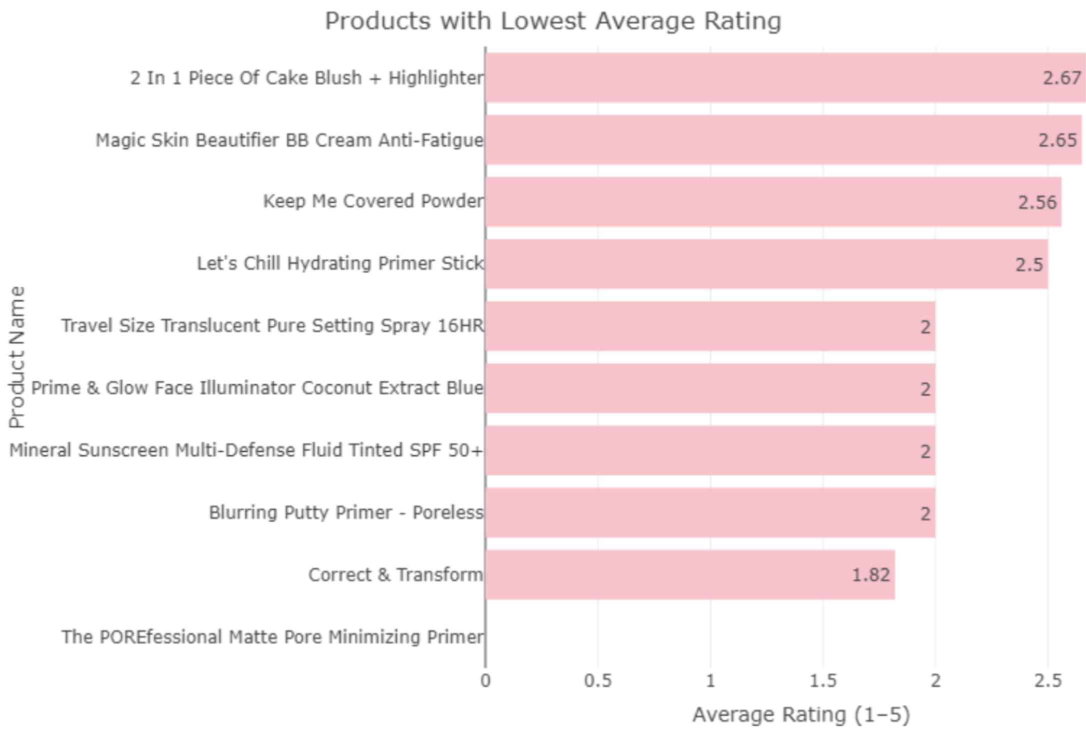

Figure 24: Makeup products with the lowest customer ratings

The bar chart presents the makeup products with the lowest average ratings in the dataset. These products have ratings ranging from 1.82 to 2.67, indicating generally poor customer satisfaction compared to the rest of the dataset. The POREfessional Matte Pore Minimizing Primer has the lowest rating at 1.82, followed by products such as Correct and Transform, Blurring Putty Primer Poreless, and Mineral Sunscreen Multi-Defense Fluid Tinted SPF 50+, each with an average rating of 2. Other low-rated items include the Magic Skin Beautifier BB Cream Anti-Fatigue and the 2 in 1 Piece of Cake Blush and Highlighter. Overall, these results suggest that while most products in the dataset receive positive feedback, a small number of items underperform in terms of quality or customer expectations.

2.20 Comparison of Average Helpful Votes by Reviewer Type

Figure 25: Makeup products with perfect customer recommendation ratios.

The dot chart displays the makeup products with the highest customer recommendation ratios. Each of the listed products has a perfect recommendation ratio of 100 percent, meaning all reviewers indicated they would recommend these products to others. These items include a mix of foundations, primers, concealers, and moisturizers such as the Mini Tinted Moisturizer Oil Free Natural Skin Perfector Broad Spectrum SPF 20, and Shape Tape Conceal and Sculpt Duo. The consistency of perfect recommendation ratios across multiple brands and product types suggests extremely positive user experiences and strong customer loyalty. It also reflects the difference in having a high average rating compared to a high recommendation ratio, with the latter reflecting consumer trust and potentially brand loyalty. This also indicates that these products are well received across different user demographics, making them top-performing items within the dataset.

2.21 Comparison of Average Product Ratings by Reviewer Type

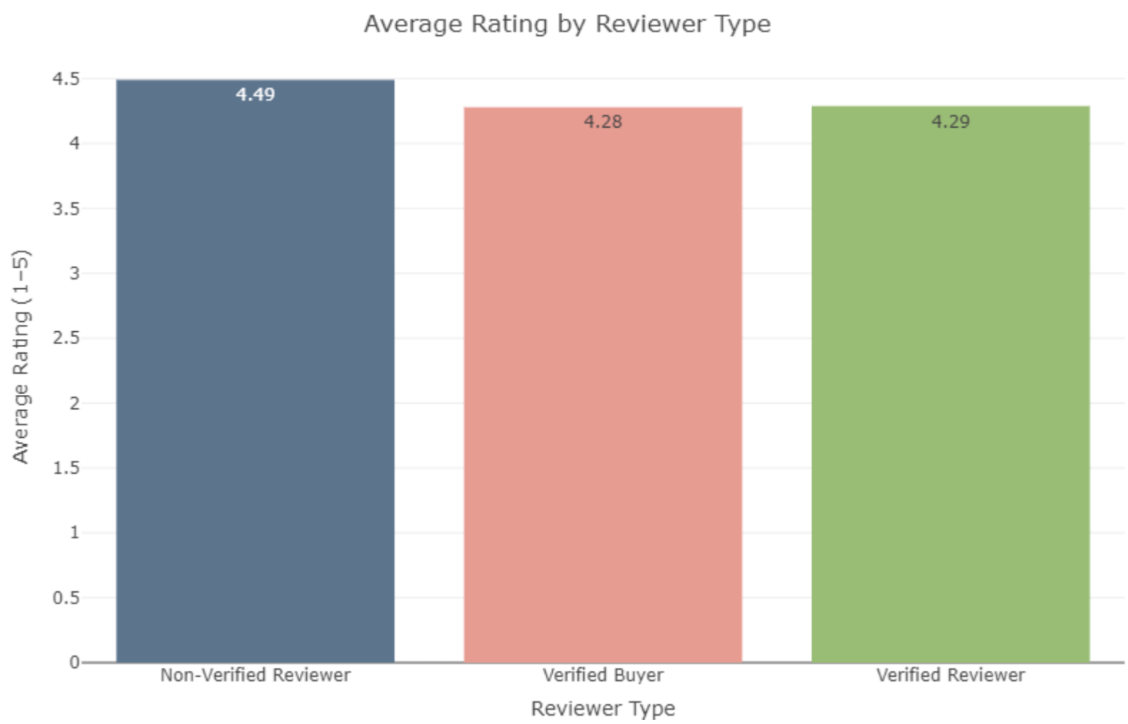

Figure 26: Comparison of average ratings across reviewer types.

The bar chart compares the average ratings given by different types of reviewers. Non-verified reviewers have the highest average rating at 4.49, followed by verified reviewers at 4.29 and verified buyers at 4.28. Although all three groups give relatively high ratings overall, non-verified reviewers tend to be slightly more generous in their evaluations. This may suggest that verified users, who have confirmed product purchases, provide more balanced or realistic feedback, while non-verified reviewers may be more likely to leave overly positive ratings. Overall, the differences between reviewer types are small but indicate potential variations in rating credibility and user behavior.

2.22 Comparison of Average Helpful Votes by Reviewer Type

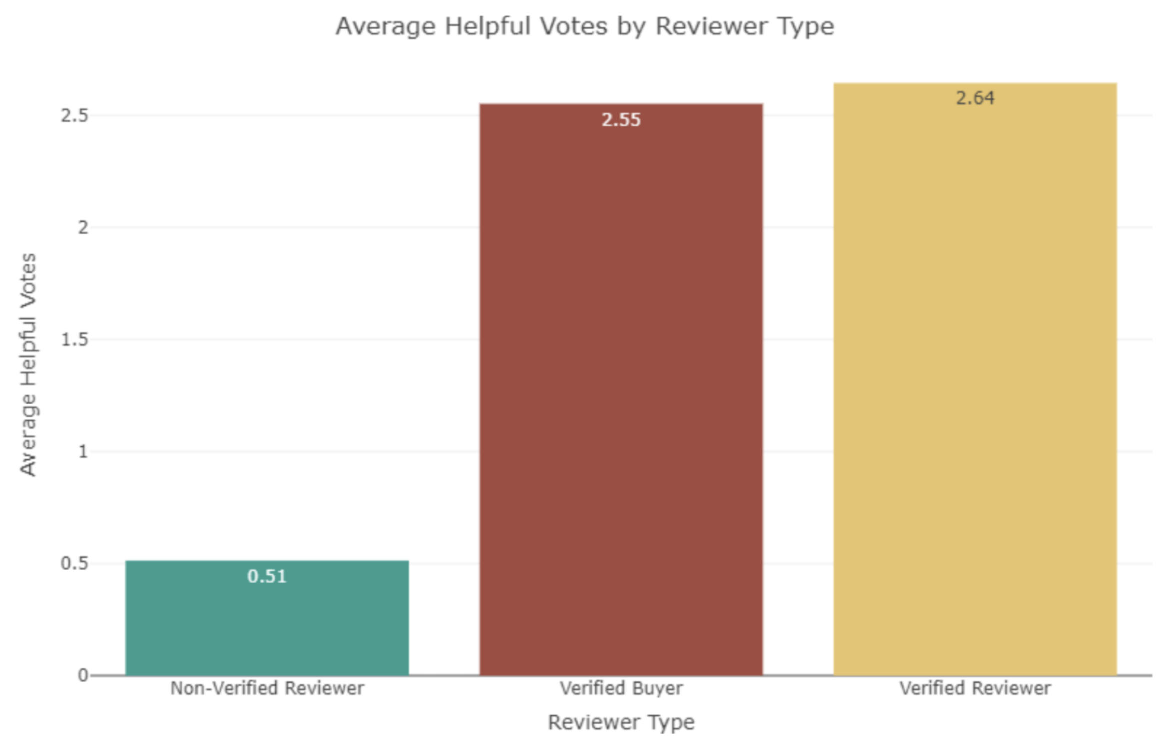

Figure 27: Comparing average helpful votes across reviewer types.

The bar chart compares the average number of helpful votes received by different types of reviewers. Verified reviewers have the highest average number of helpful votes at 2.64, followed closely by verified buyers at 2.55. Non-verified reviewers receive the fewest helpful votes, averaging only 0.51. This pattern suggests that reviews from verified users are perceived as more credible and useful by other customers, likely because they are based on confirmed purchases or genuine product experiences. Overall, verified reviews appear to hold greater trust and influence within the review community compared to non-verified feedback.

2.23 Average Rating by Combined Reviewer Types

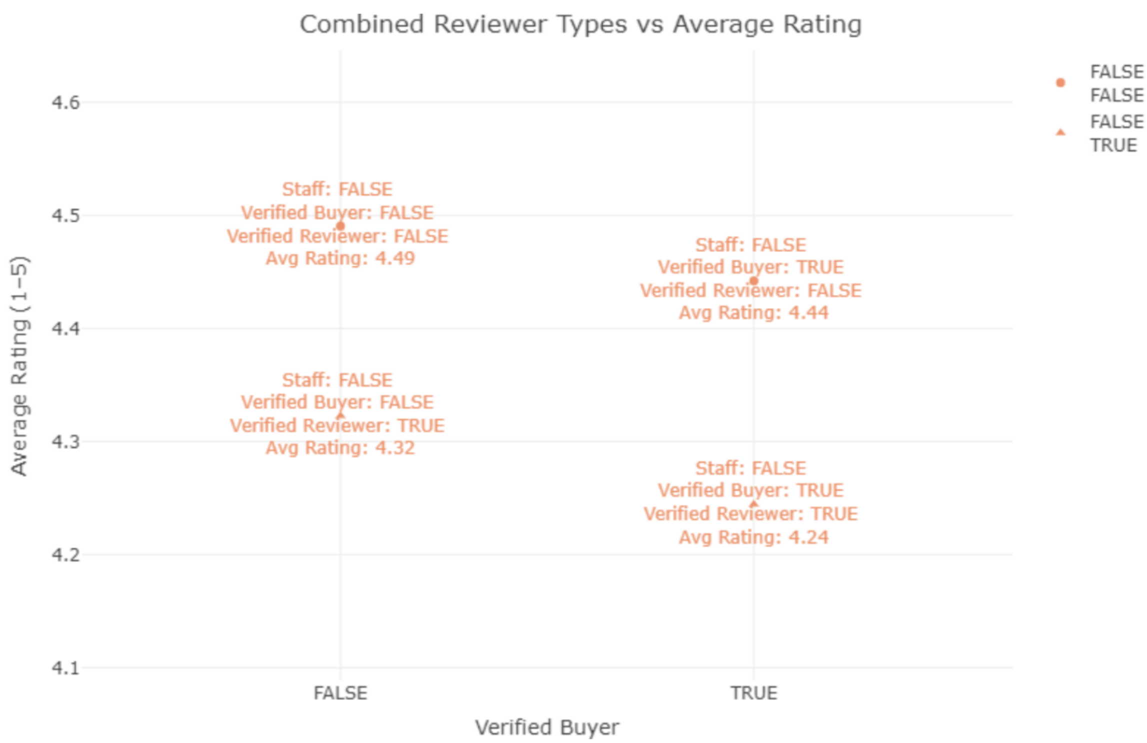

Figure 28: Average rating by combinations of reviewer verification attributes.

The scatterplot shows how different combinations of reviewer types relate to the average rating given to makeup products. The plot compares verified buyers and verified reviewers, including whether the reviewer is staff or not. The highest average rating of 4.49 comes from non-verified buyers and non-verified reviewers, while verified buyers and verified reviewers together have slightly lower average ratings around 4.24 to 4.32. Verified buyers who are not verified reviewers show an average rating of 4.44. Overall, non-verified reviewers tend to give slightly higher ratings than verified ones. This pattern suggests that verified reviewers and buyers may provide more balanced and possibly more critical feedback, whereas non-verified reviewers often rate products more positively.

References

[1] Z. Sarkar. Luxxify: Ulta makeup products. https://www.kaggle.com/datasets/zusmani/ulta-makeup-products-reviews, 2024. Kaggle dataset.

[2] Z. Sarkar. Luxxify: Ulta makeup reviews. https://www.kaggle.com/datasets/zusmani/ulta-makeup-products-reviews, 2024. Kaggle dataset.